



Amazon with Arianespace, Blue Origin and United Launch Alliance will deploy thousands of satellites into low-Earth orbit (Source: <https://www.cnet.com>)



Jio Satellite Communication Ltd has a licence for global mobile personal communications by satellite services (Source: <https://www.desidime.com>)

of the company not to take pre-bookings before getting the licence or authorisation from them.

Satellite internet for the US Armed Forces

When terrestrial internet connections are disabled or blocked by enemy action, satellite internet provides an alternate for effective support. Recognising the importance, the US Space Development Agency (SDA) has signed a dual-use contract with Starlink to develop a military version to provide the US military capability to dominate the cyberspace.

Starlink has started modifying the systems to make them more ruggedised, improve their compatibility with the current satcom equipment, and equip the satellites with devices to provide combat capabilities for reconnaissance, remote sensing,

communications relay, navigation, and positioning. As part of the project, Starlink has located some of the launch sites in the US Air Force bases.

The first tranche of satellites forming part of the Space Force's National Defense Space Architecture (NDSA) has been launched

with capabilities for data transport, battle management, missile tracking, weapons targeting, navigation, deterrence, and ground support. The system has been successfully connected to a variety of air and terrestrial assets of the US to establish encrypted interconnections between the satellites and the US Air Force fighter planes.

Satellite internet for the Indian Army

The Indian Army has been studying the ongoing cyber and electromagnetic aspect of the Russia-Ukraine war and the advancements made by the adversaries in the field. The study has established the need for having a tactical high-speed satellite internet to support military operations and communications, especially in the remote areas.

Currently, the Indian Army does not have a dedicated satellite for

this purpose. However, the Defence Acquisition Council has cleared a proposal for an advanced satellite, GSAT-7B, which is being built by ISRO and is likely to be available by December 2025.

Using the existing infrastructure, the Indian Army has activated satellite-based internet service on Siachen Glacier at 5,810 metres—the world's highest battlefield. Bharat Broadband Network Limited (BBNL) has a special purpose vehicle (SPV) set up by the government to create national optical fibre network pan India to cover around 250,000 *gram panchayats* spread over 641 districts, the borders, and the other operational areas for providing the net connectivity.

BBNL is acquiring four satellite gateways with two high-powered satellites to provide inter-BharatNet pan India. In addition, a few private companies are providing satellite based internet connectivity to the Indian Army in sensitive places along the border. For example, Hughes Communications India is providing satellite based connectivity along the China border, including the Galwan region.

Role during Russia-Ukraine war

On February 26, 2022, two days after the Russian invasion, the Ukrainian army suffered heavy losses due to signal towers' and underground communication cables' damage,

thus disrupting the internet and phone connections. Ukraine requested SpaceX to provide Starlink terminals. SpaceX deployed 10,000 satellite terminals to restore the destroyed internet connections of military bases, hospitals, and homes.

The Ukraine army has deployed the



Starlink is providing the US military capabilities to dominate the cyberspace (Source: <https://www.airandspaceforces.com>)